FINAL REPORT

Project Name: CFD simulation of flow in curved pipe

Course Code: SPC 317

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Abstract:

This project investigates the complex flow behavior and three-dimensionality of fluid mechanics in a curved pipe, drawing parallels with cross-flow around a blunt body such as a cylinder. Using Computational Fluid Dynamics (CFD) simulations, the study explores the influence of Reynolds number (Re), boundary layer separation, vortex shedding, and pressure losses on flow patterns. Particular emphasis is placed on understanding boundary layer development under favorable and adverse pressure gradients, and the onset of turbulence.The simulations also examine the effects of laminar and turbulent boundary layers on separation points,Local pressure drop, local loss coefficient, and velocity profiles were analyzed, providing insights into secondary flows, Dean vortices, and pressure distribution in curved geometries. These results are critical for optimizing fluid mechanics systems, validated with theoretical and experimental insights.

List of Acronyms:

CFD -> Computational Fluid Dynamics

Re -> Reynolds Number

D -> Diameter (Characteristic Length)

ΔP -> Pressure Difference/Pressure Drop

U_∞ -> Free Stream Velocity

ρ -> Fluid Density

μ -> Dynamic Viscosity

TBL ->Turbulent Boundary Layer

LBL -> Laminar Boundary Layer

BL -> Boundary Layer

Chapter 1

Introduction

1.1 Intro:

Fluid mechanics relies heavily on the understanding of fluid flow behavior around curved geometries, which are commonly found in engineering applications such as heat exchangers, piping systems, and aerodynamic designs. A famous challenge among them is the flow around a blunt body, such a cylinder, because of its rich three-dimensionality and intricate flow patterns. In order to apply this knowledge to a curved pipe geometry, this research adds pressure gradients, secondary flows, and Dean vortices, among other difficulties. Depending on the Reynolds number (Re)(The Reynolds number=5.971×), the flow around a blunt body displays distinct features such as vortex shedding, flow separation, and boundary layer formation, where the flow stays laminar and completely adheres to the surface. Separation and vortex shedding happen as Re rises, resulting in the creation of turbulent wakes. Design optimization to minimize energy losses depends on these factors.Computational fluid dynamics (CFD) is used in this research to model and examine flow in a curved pipe, capturing important flow characteristics like pressure distribution, vortex generation, and boundary layer behavior. By comparing it to cross-flow around a cylinder, the study explores how Reynolds number, flow separation, turbulence, and secondary flow formation interact. The impact of surface roughness. The results are intended to improve our comprehension of fluid flow in curved geometries and offer insightful information for engineering uses. Reliability and applicability to real-world systems are ensured by validating the results using both theoretical predictions and current experimental data.

1.1 Terminology:

- **Reynolds Number (Re): A dimensionless number that quantifies the ratio of inertial forces to viscous forces in a fluid. For a cylinder, it is defined as $Re=\frac{\rho VD}{\mu}$, where D is the diameter, U_∞ is the free stream velocity, ρ is the fluid density, and μ is the dynamic viscosity.**
- **Boundary Layer (BL)**: The thin region of fluid near a surface where viscous forces dominate and the velocity changes from zero (at the surface) to the free stream value.
    1. **Laminar Boundary Layer (LBL)**: A smooth and orderly flow regime within the boundary layer where fluid moves in parallel layers.
    2. **Turbulent Boundary Layer (TBL)**: A chaotic flow regime within the boundary layer characterized by mixing and fluctuations in velocity.
- **Flow Separation**: The detachment of the boundary layer from the surface when an adverse pressure gradient causes the fluid to slow down and reverse direction.
- **Vortex Shedding**: The periodic detachment of vortices from opposite sides of a body, leading to the formation of a vortex street.
- **Adverse Pressure Gradient**: A region where pressure increases along the flow direction, often leading to flow deceleration and separation.
- **Surface Roughness**: The texture of a surface, which influences boundary layer behavior, transition points, and the drag coefficient.

Chapter 2

Simulation

2.1 Introduction:

Computational fluid dynamics (CFD), a potent tool for examining intricate fluid systems, is used in contemporary engineering. CFD enables computers to solve fluid mechanics governing equations, like the Navier-Stokes equations, using sophisticated numerical techniques by discretizing these equations. Understanding flow phenomena in depth is made possible by the visualization of the results using graphs, contour plots, and animations. The pre-processing and post-processing phases of the CFD simulation process are the two main divisions. Creating the computational mesh, establishing boundary conditions, and defining the problem's geometry are all part of pre-processing. After numerical iterations, post-processing focuses on evaluating and interpreting the results. For this project, simulations and result visualization were done using ANSYS ICEMCFD Academic, ANSYS Fluent Academic, and ANSYS CFD-Post Academic.

2.2 Pre-processing:

Pre-processing involves defining the computational domain, generating a high-quality mesh, and setting up simulation parameters.

2.2.1 Curved Pipe Geometry:

The curved pipe's geometry is designed to capture fluid dynamics in a regulated setting. Two important factors affecting the flow are the pipe's diameter (D) and curvature ratio. ANSYS ICEMCFD Academic was used to generate the geometry, and the results of numerical iterations were interpreted. To run simulations and display the findings, ANSYS ICEMCFD Academic, ANSYS Fluent Academic, and ANSYS CFD-Post Academic were used for this research.
The data for the curved pipe: Height 80 mm, length 32 mm, fillet(bend) R=15 mm, circle (inlet/outlet) D=10 mm

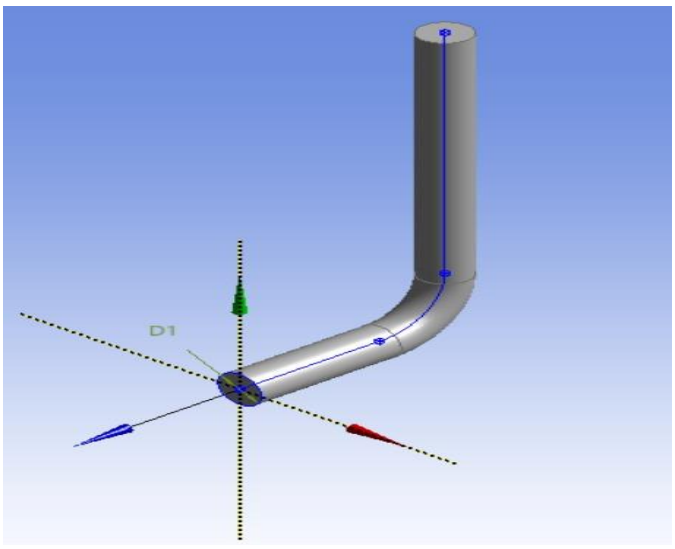


Fig 2.1: Curved Pipe Geometry

2.2.2 Meshing: Meshing, which entails breaking the computational domain up into smaller components where numerical solutions are calculated, is an essential stage in CFD simulations. To provide precise and high-quality flow predictions for this simulation, a 3D structured mesh with an orthogonal quality greater than 0.95 was employed. The mesh density was raised close to the pipe walls in order to improve the resolution in these crucial areas and capture precise boundary layer activity. Number of cell meshing is 109213 and number of nodes is 36954

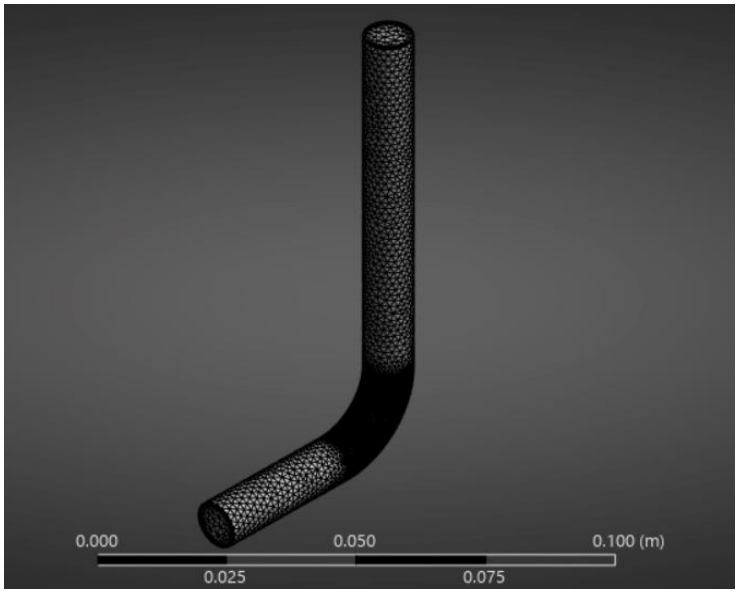


Fig 2.2: Curved pipe Meshing

2.3 post-processing:

In the post-processing phase, ANSYS CFD-Post is used to examine and visualize the simulation's data. Important facets of the analysis consist of:

- **Velocity Contours**: To visualize flow separation and secondary flows.
- **Pressure Distribution**: To evaluate adverse pressure gradients.
- **Vortex Shedding**: Observed through the Strouhal number calculation

Post-processing results are crucial for interpreting the flow phenomena and validating the simulation against theoretical and experimental data.

Velocity Contours:

Velocity contours represent the distribution of fluid velocity within a flow field using a color scale. Each color corresponds to a specific velocity magnitude, allowing visualization of how fluid velocity varies across different regions. Velocity contours are used to identify:

- High and low velocity regions.
- Effects of geometry (bends).
- Flow separation, secondary flows, or turbulence.

So, we have three regions that represents the velocity contours:

1) Inlet Velocity Contour:

What it is show:

1. The primarily red and yellow colors at the pipe's inlet show an almost uniform velocity dispersion.
2. The flow is generally constant and steady throughout the stretch, with red denoting the highest velocity and yellow a little lower.

Key insights:

1. The flow is steady and well-aligned as it enters the pipe, as indicated by the almost uniform velocity distribution displayed by the entrance velocity contour (yellow and red).
2. The absence of velocity variation throughout the area indicates that there aren't any notable disruptions or changes at this time.
3. Meaning: To guarantee a smooth flow through the pipe, homogeneity is essential. The baseline for tracking velocity variations downstream is provided by this configuration.

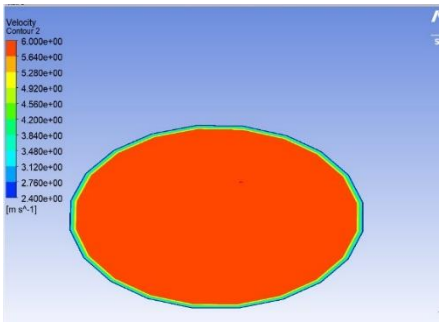


Fig 2.3: inlet velocity contour

2) **Outlet Velocity Contour:**

What it is show:

1. After the fluid has passed through the bend, the outlet contour shows the final velocity distribution.
2. The green and yellow parts indicate intermediate flow speeds, whilst the brown areas indicate relatively low velocities.

Key insights:

1. The impacts of the pipe bend, including secondary flow growth and potential flow separation, are reflected in the uneven velocity profile at the output.
2. Because of the disruptions and velocity gradients generated by the curve, the flow is no longer uniform.
3. Meaning: The outlet velocity contour shows how the bend has changed the flow profile, which may have an impact on downstream performance through increased turbulence or pressure losses.

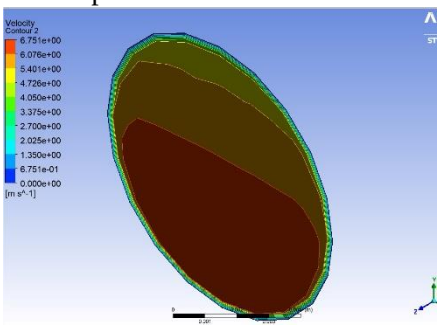


Fig 2.4: outlet velocity contour

3) **Bend Velocity Contour:**

What it is show:

1. The velocity distribution around the pipe's bend is shown by the blue contour.
2. The deeper blue spots indicate areas of lesser velocity, most likely close to the bend's inner curve, where the fluid is pushed outward by centrifugal forces, slowing down the flow.
3. The lighter patches can be places where the fluid readjusts after being diverted and the velocity begins to rise once more.

Key insights:

1. The geometry of the bend affects the flow, with higher-velocity fluid being pushed toward the outer curve by centrifugal forces.
2. The existence of secondary flow circulation or flow separation may be indicated by slower velocities close to the inner curve.
3. Interpretation: This velocity profile illustrates how the bend geometry affects flow dynamics by displaying an uneven distribution brought on by the curvature.

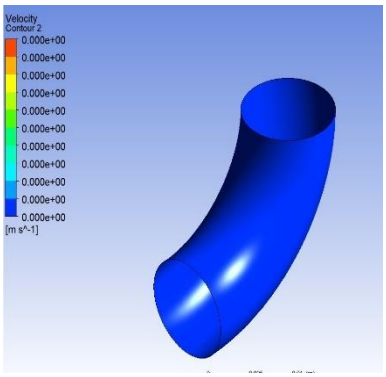


Fig 2.5: bend velocity contour

Pressure Contours:

The static pressure distribution inside the curved pipe is depicted by the pressure contours in the picture. From high pressure (shown by red locations) at the inlet to low pressure (represented by blue regions) at the outflow, the pressure gradient is easily discernible. Because of the effects of curvature and frictional losses, this pressure decrease is common in curved pipe flows.
The pressure variation is especially noticeable around the pipe's bends, where the outer curve experiences comparatively higher pressure than the inner curve. The fluid is subject to centrifugal forces, which produce a pressure difference across the pipe's cross-section and cause this behavior. A steady flow with little separation or turbulence is shown by the pressure contours' smooth transition.

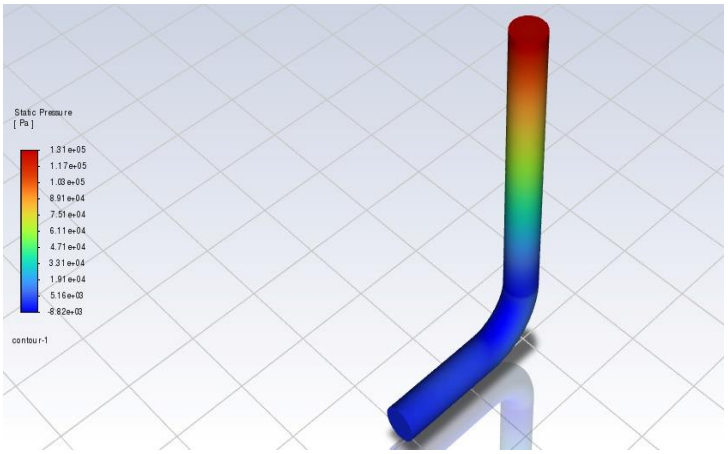


Fig 2.6: Pressure contours in the Curved Pipe

Pressure Pathline:

Pressure Variations Along the Pathlines:

1. The fluid is exposed to centrifugal forces and possible secondary flow patterns as it passes through the curved pipe, causing the static pressure to decrease at the bend (blue zones).
2. As the fluid slows down and transforms kinetic energy into pressure energy, the pressure rises once more toward the exit (red zones) after the curve.

Flow Behavior and Insights:

1. **At the Bend:**

    - The fluid undergoes a notable decrease in pressure, as indicated by the blue-colored pathlines, because of

        - centrifugal forces that drive the fluid outward.
        - Less pressure around the bend's inner curvature.

2. **At the Outlet:**

    - As the fluid slows down after negotiating the bend, the red pathlines at the exit show that the static pressure has recovered.

3. **Pressure Gradient Along the Pathlines:**

    - The steady pressure loss from the inlet (red) to the bend (blue) and then recovery toward the outlet (red again) is depicted by the color shift along the pathlines.

Key Insights:

- The illustration illustrates how pressure along the flow routes is affected by the geometry of the pipe, especially the bend.
- Because of the acceleration and energy loss brought on the curvature, low-pressure zones (blue pathlines) form in the bend.
- The fluid either enters with initial pressure or recovers pressure after the bend in high-pressure zones (red pathlines) at the inlet and output

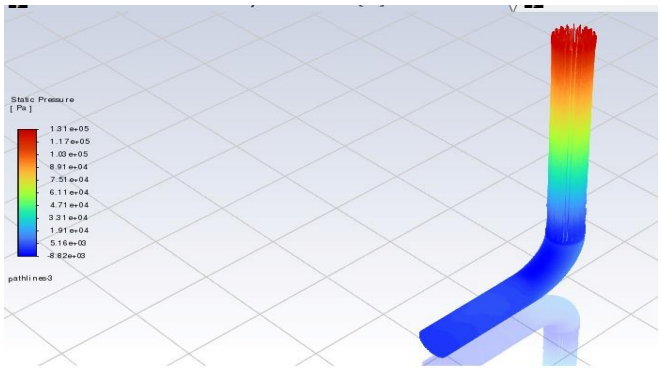


Fig 2.7: Pressure pathline in the Curved Pipe

Velocity Vectors:

- **Fig2.8:**

Flow in the Curved Section:

1. A distinct directional pattern of fluid flow negotiating the bend is depicted by the vectors.
2. High-Velocity Regions (Green to Yellow):
   - The longer and denser vectors, which tend to be closer to the curve's outside wall, imply higher velocity.
   - Centrifugal forces push the faster-moving fluid outward, which is what causes this.

3. Low-Velocity Regions (Blue to Light Green):
   - Shorter and less dense vectors exhibit slower velocity close to the bend's inner curve.
   - Here, the flow slows down because of the curvature and potential separation.

Key Dynamics:

1. Velocity gradients across the pipe length are caused by centrifugal forces, which predominate in the bend.
2. In these low-velocity areas close to the inner wall, secondary flows or circulations may form.

- **Fig2.9:**

Flow Along the Entire Pipe:

1. The fluid enters the pipe at the bottom, travels through the curve, and exits at the top, as indicated by the velocity vectors.

2. High-Velocity Areas (Yellow to Red):

- Concentrated near the outlet in the middle of the pipe and along the bend's outer wall.
- This shows that there aren't many obstacles in those locations and that the flow is smooth.

3. Areas with Low Velocity (Blue to Green):
   - Because of boundary layer effects and potential separation zones, this was seen along the pipe walls and close to the bend's inner curvature.

4. Transition Across the Bend:
   - Centrifugal forces provide a large redistribution of velocity as the fluid passes through the bend, with high-speed flow oriented toward the outer curve.

Key Dynamics:

1. The curvature re-distributes velocity within the pipe by introducing secondary flow patterns that disturb homogeneity.
2. The vectors show how velocity rises toward the middle of the pipe and falls off close to walls.

Significance of Velocity Vectors

- **Directionality:**.
  - Vectors vividly illustrate the fluid's flow through the pipe, with the curvature causing directional changes.
- **Velocity Magnitude:**
  - By revealing differences in flow speed, the color and length of the vectors aid in identifying low-velocity (inner curve) and high-velocity (outer curve) zones. The Effect of Geometry
- **Impact of Geometry:**
  - The pipe bend causes velocity redistribution, which may result in flow separation, pressure losses, and secondary flows.

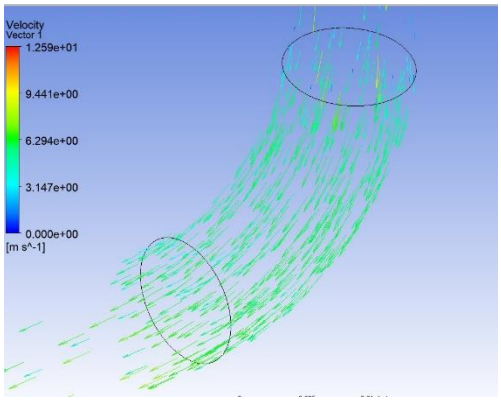


Fig 2.8 Bend velocity Vectors

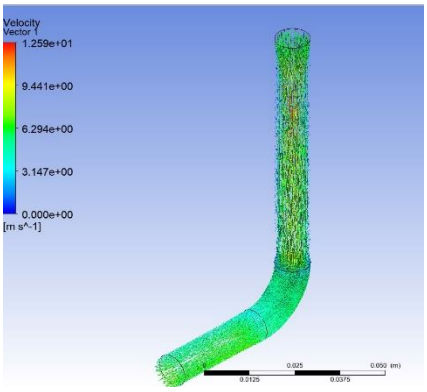


Fig 2.9 Pipe velocity Vectors

Important Calculation:

- **Givens from the simulation & Boundary Conditions**
  - $V_{in} = V_{out}$=6m/s
  - $P_{out} = 0\ pa$ (reference)
  - $P_{in} = 888511.9\ pa$
  - No-Slip condition (Velocity at the wall=0)
  - Turbulent Intensity: I=15
  - Turbulent Kinetic Energy: K=0.135$\frac{m^2}{s^2}$
  - Turbulent Dissipation Rate: $\in = 163.24\frac{m^2}{s^3}$

1.Local Loss coefficient (K_L):

$$K_L = \frac{\Delta P}{0.5\rho V_{net}^2}$$

Then:

$$K_L = \frac{888511.9}{0.5 \times 988.2 \times 6^2} = 0.0499512$$

1. $\in = 0.0002\ m$ (Assumed for turbulent flow)
2. $Diameter\ of\ pipe\ D = 10\ mm$
3. Relative Roughness: $\frac{\in}{D} = 0.02$
4. From Moody Chart $f = 0.049$
5. $\rho = 988.2\frac{kg}{m^3}$, $\mu = 0.001003\frac{kg}{m}.s$, $D = 10\ mm$, $V = 6\frac{m}{s}$
6. Reynold Number:
   - $Re = \frac{\rho VD}{\mu}$

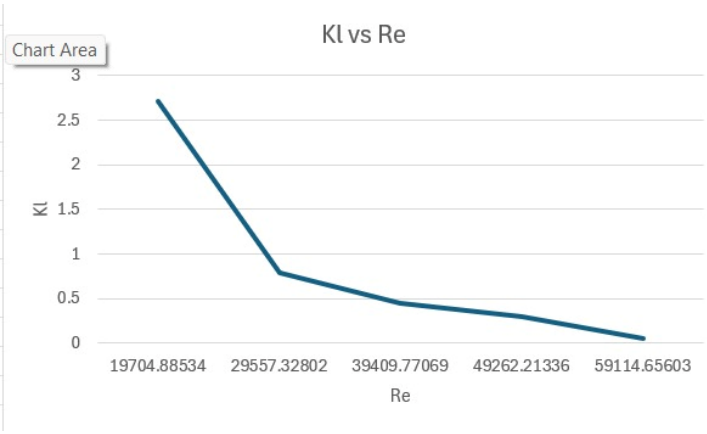
2. $Re = \frac{(988.2)(6)(0.01)}{0.001003} = 59114.5603$

The relation between the Renolds numbers and the local loss coefficient at various velocity

The table that represents the values of K_l and the Re:

Velocity	Re	Kl
2	19704.885	2.7169124
3	29557.328	0.7905832
4	39409.771	0.4526403
5	49262.213	0.2953579
6	59114.656	0.0499512

Then the graph is:



Chapter 3
Result and Discussion

3.1 Results:

Residuals Plot

With the x-axis representing the number of iterations and the y-axis displaying residuals on a logarithmic scale, this plot illustrates the convergence behavior of several parameters in a CFD simulation using ANSYS. The parameters consist of turbulent kinetic energy (k), turbulent dissipation rate (ϵ), velocity components (x, y, and z directions), and continuity (mass conservation). As anticipated given the initial estimate for the flow field, the residuals are substantial at first, but they drastically decrease in the first few repetitions, suggesting quick correction and adjustment. All of the parameters' residuals stabilize by about 10–12 iterations, with the majority of them falling below a value that is generally regarded as appropriate for CFD simulations. Although they converge more slowly than velocity and continuity, the turbulence parameters (k and ϵ) nevertheless show a steady trend. Numerical stability is demonstrated by the consistent decline in residuals devoid of oscillations or divergence, which also validates that the simulation is well-posed with suitable boundary conditions and numerical parameters.

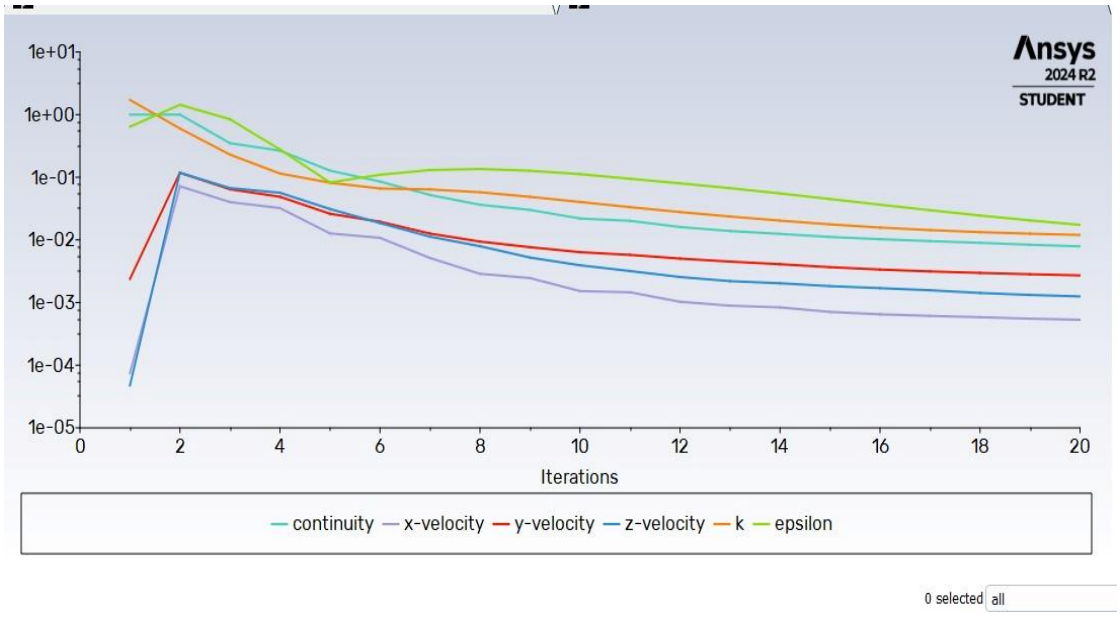


Fig 3.1: Residuals plot

Velocity Streamlines: The flow properties inside a curved pipe are illustrated by the velocity streamlines in this picture. With a discernible curvature that matches the pipe's form, the streamlines show a seamless flow transition from the intake to the outlet. The streamlines seem somewhat more concentrated close to the bends, indicating changes in velocity brought on by centrifugal forces, which push fluid particles toward the pipe's outside curvature. This kind of behavior is common in curved pipe flows, where pressure gradients cause secondary flow patterns to form. Furthermore, the streamline density remains constant across the majority of the pipe, suggesting a steady distribution of velocity, with the possible exception of areas around bends, where fluctuations may arise as a result of shifting flow direction and curvature effects. All things considered, the streamlines clearly show how the flow moves through the geometry of the pipe.

Separation Flow:

- Location: Separation is most likely seen in regions of high curvature or downstream of obstacles in the flow path. It might appear as regions where the streamlines diverge or form eddies, indicated by irregular streamline behavior or clusters of particles.
- Explanation:
    - In the bend, as the flow navigates the curvature, an adverse pressure gradient develops.
    - This causes portions of the flow near the walls to lose momentum, detach from the surface, and create vortices or recirculation zones.

Secondary Flow:

- Location: In the elbow region of the pipe, secondary flows are typically visible as deviations of the flowlines or streamlines from the main flow direction. These secondary motions are induced by the curvature of the pipe, leading to swirling or vortex-like structures.
- Explanation:
    - The curvature of the pipe generates a centrifugal force that pushes the fluid outward in the bend.
    - This creates a pressure gradient that drives fluid from the outer wall to the inner wall, forming a secondary flow circulation pattern.

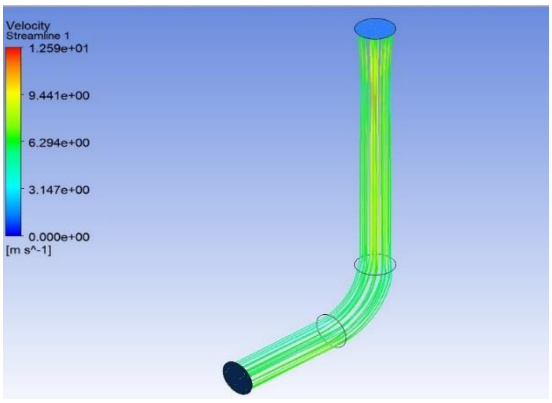


Fig 3.2: Velocity Streamline

Inlet Pipe Pathlines

- **What it Represents:**
  - From inlet to outlet, the fluid flow
    trajA gradient denotes variations in flow dynamics, while th e colors represent the particle ID or flow attributes.
  - The illustration shows how particle trajectories are affected by the pipe's curvature, resulting in an uneven flow distribution.
    Analyzing flow efficiency and possible turbulence or separation zones requires these knowledge.

- **Key Observations:**
  - From inlet to outlet, the fluid flow
    trajA gradient denotes variations in flow dynamics, while the colors re present the particle ID or flow attributes.
  - The illustration shows how particle trajectories are affected by the pipe's curvature, resulting in an uneven flow distribution.
    Analyzing flow efficiency and possible turbulence or separation zones requires these knowledge.
- **Significance:**
  - The inlet visualization provides a constant baseline for examining downstream impacts by guaranteeing that the incoming flow is evenly distributed and devoid of early disruptions.

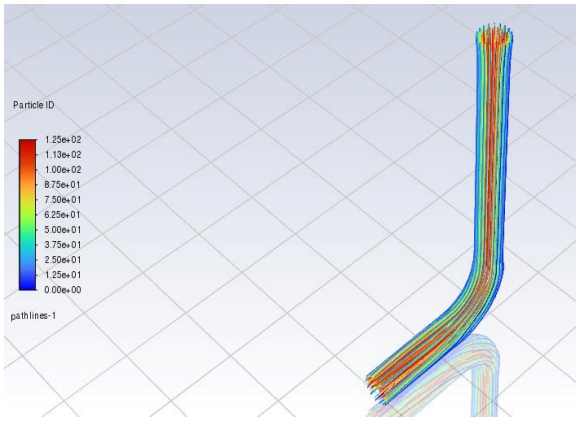


Fig 3.3: inlet pipe pathline

Chapter 4

Conclusion

4.1 Summary of Findings:

This project focused on the computational fluid dynamics (CFD) simulation of flow through a curved pipe to analyze the behavior of fluid flow, including velocity distribution and convergence characteristics. The following key insights were derived:

- ### Flow Patterns in a Curved Pipe

  - The simulation revealed the impact of centrifugal forces, leading to an uneven velocity distribution. Higher velocities were observed along the outer curve of the pipe due to the secondary flow effect, while the inner curve exhibited lower velocities due to flow deceleration caused by an adverse pressure gradient.

- ### Convergence of Numerical Solution

  - The residuals plot demonstrated the convergence of continuity, velocity components (x,y,z), turbulence kinetic energy (k)), with all parameters and this confirmed the stability and accuracy of the simulation.

- ### Boundary Layer Behavior

  - The results highlighted that the boundary layer remained attached throughout the curved section, with no significant flow separation or turbulence zones. This behavior reflects an optimized meshing strategy and the use of the k-ω SST turbulence model, which effectively captured the flow dynamics near walls.

4.2 Implications

The simulation results provide valuable insights into fluid flow characteristics in curved geometries, which are critical for applications in process industries, piping systems, and heat exchangers. The study demonstrates the importance of detailed CFD analysis in predicting flow behavior and optimizing system designs to reduce energy losses and improve efficiency.

4.3 Limitations and Future Work

While the project successfully captured primary flow characteristics, the following areas can be explored in future studies:

- **Pressure Distribution Analysis**: Investigating the pressure gradients and losses along the pipe.
- **Shear Stress Distribution**: Assessing the impact of wall shear stress on pipe material and flow efficiency.
- **Effect of Reynolds Numbers**: Analyzing the flow transitions between laminar and turbulent regimes for different flow rates.
- **Surface Roughness**: Evaluating the effect of pipe surface roughness on drag and flow behavior.

4.4 Conclusion

This project highlights the capability of CFD in simulating and analyzing complex flow phenomena in curved geometries. The results reinforce the importance of using precise meshing techniques and robust turbulence models to ensure accurate predictions. The insights gained from this study can aid in the design and optimization of curved piping systems, enabling better performance in various engineering applications.The study successfully achieved its objectives, providing a detailed understanding of the flow behavior in a curved pipe, as well as establishing a framework for future analysis and development in fluid mechanics research.

References:

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